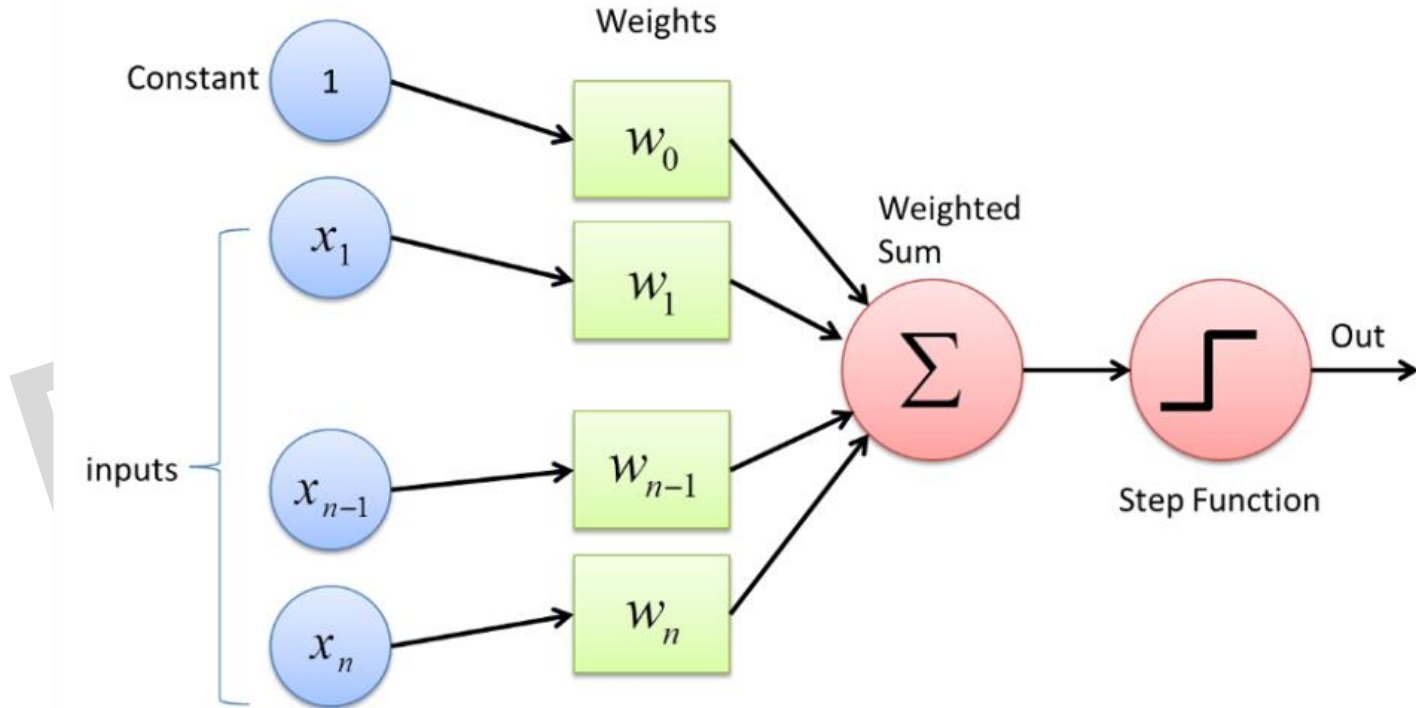


Deep Neural Network

Project Pro

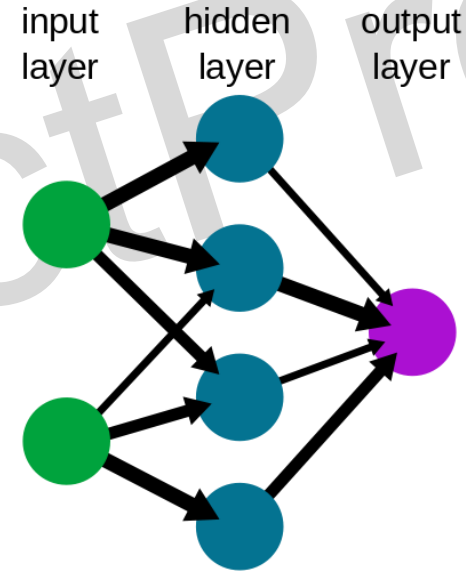
Perceptron



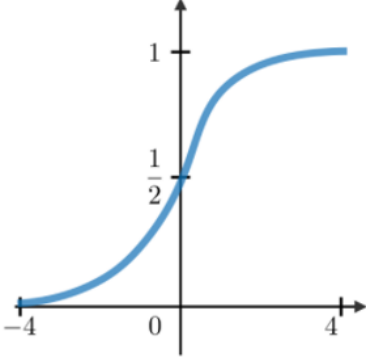
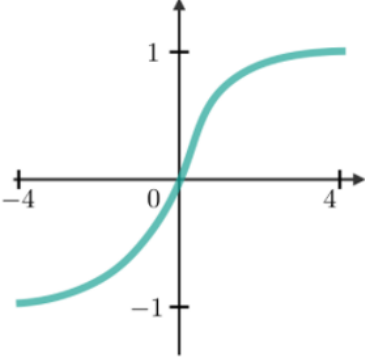
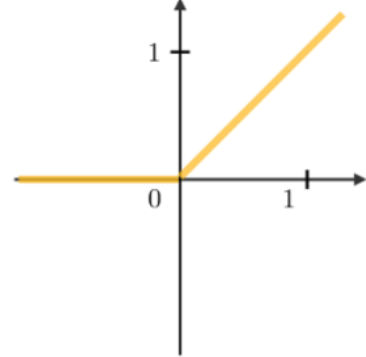
Perceptron -> Neural Network

- Input Layer
- Nodes
- Hidden Layer
- Activation function
- Output Layer

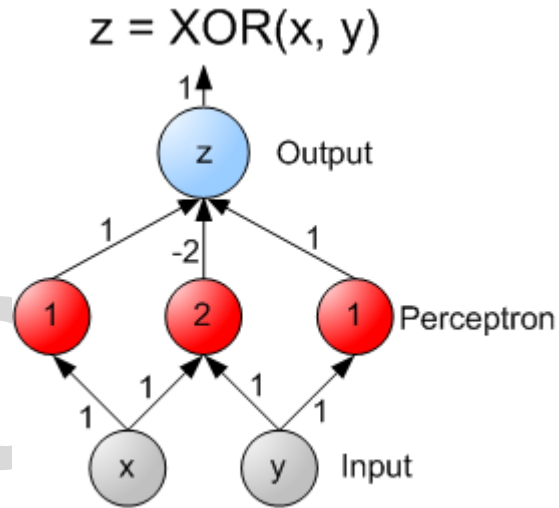
A simple neural network



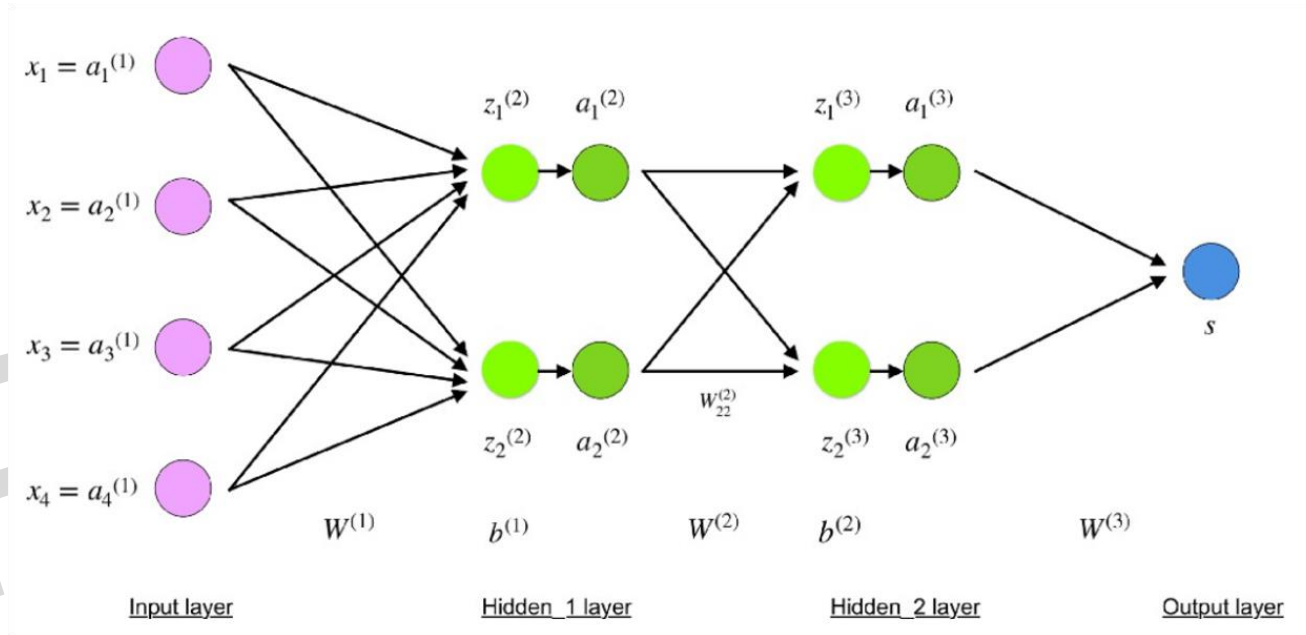
Activation Function

Sigmoid	Tanh	RELU
$g(z) = \frac{1}{1 + e^{-z}}$	$g(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$	$g(z) = \max(0, z)$
		

Feedforward



Backpropagation



Loss Function

- Mean Absolute Error

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n}$$

- Mean Squared Error

$$MSE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}$$

Loss Function

- Cross-entropy Loss

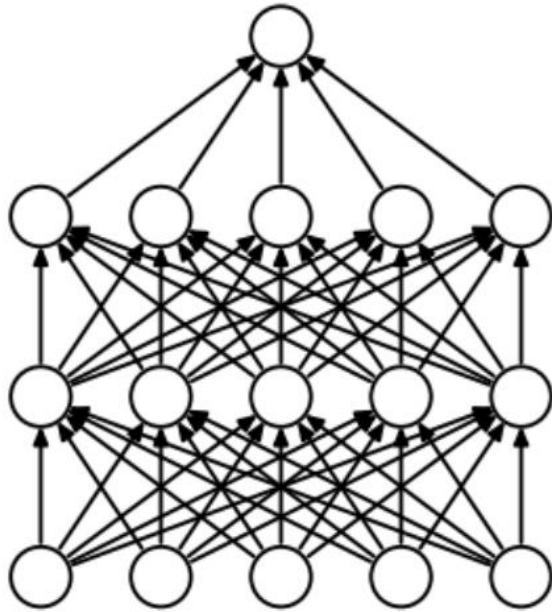
$$CrossEntropyLoss = -(y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i))$$

Cross Entropy

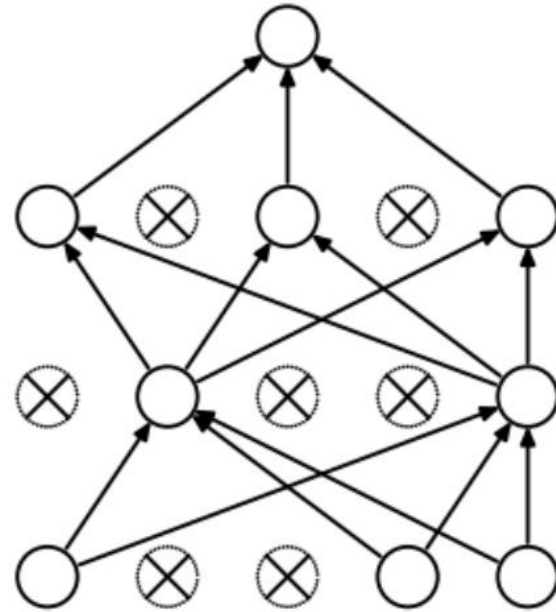
- Hinge Loss

$$SVMLoss = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

Regularization - Dropout



(a) Standard Neural Net



(b) After applying dropout.